# Cycle Survival Under Poisson Perturbation of Random Permutations

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Originally released 2026; the general-$K$ rate, Open Lemma 4, and Proposition 5 became unconditional on 2026-08-22 – see the note after the abstract.

**Abstract**

Let $\pi$ be a uniformly random permutation of $[n] = \{1, \ldots, n\}$, and independently reroute each point $i$, with probability $c/n$, to a uniform destination in $[n]$ (leaving $f(i) = \pi(i)$ otherwise). We give a closed form for the $n \to \infty$ limit of the expected fraction of points left on a cycle of $f$:

$$\varphi_\infty(c) \;=\; \int_0^1 e^{-ct^2}\,dt \;=\; \tfrac{1}{2}\sqrt{\pi/c}\,\operatorname{erf}(\sqrt{c}),$$

proved directly on the explicit $n \to \infty$ limit object (an exploration-process argument with an exact cancellation), together with its entire-function series, a tail asymptotic with a rigorous exponentially-small error bound, and the law of the cyclic mass conditional on exactly $K$ surviving reroutes. That conditional law's mean is proved for every $K$ and coincides, for a structurally different microscopic model, with a theorem of Hansen and Jaworski [5]. The finite-$n \to \infty$ bridge is proved in full, unconditionally, for $K = 0, 1, \ldots, 10$ – by two structurally different techniques, a hand case-analysis at $K = 1, 2$ and a $K$-uniform transfer-matrix/Markov-chain method, run mechanically through ten levels, at $K = 3, \ldots, 10$ – with exact rates $O(1/n^2)$ at $K = 1$ and $\Theta(1/n)$ at $K = 2, \ldots, 10$, not the naively extrapolated $O(1/n^2)$; the conjectured general-$K$ rate $\lim_n n(\psi_n^{(K)} - \varphi_K) = K\varphi_K/4$ is confirmed unconditionally at all eleven of these values directly from the exact closed forms, and is proved for *every* $K$ by a new continuum scaling-limit argument.[1] At the time of that proof, it carried one explicitly named, adversarially-scrutinized conditional hypothesis (a regularity/existence assumption on an asymptotic expansion, found to hold without exception at every one of the eleven checkable values); for $K \geq 11$ the fixed-$K$ bridge itself, $\varphi_n^{(K)} \to \varphi_K$, was at that time open unconditionally (though it was, like the rate, discharged conditionally on the same hypothesis, as an immediate corollary of the rate proof), via an unconditionally proved mixing lemma reducing the full statement to that single precisely stated convergence lemma. Every claim below is labeled by its exact status; a companion numerical package cross-checks every closed form against independent brute-force enumeration, Monte Carlo simulation, and quadrature.

**Note added, 2026-08-22.** Hypothesis 1 below – the sole caveat carried by Theorem 2 and Open Lemma 4 for $K$ beyond the eleven concretely verified values – has since been proved unconditionally, for every $r \geq 0$, by a discrete-Gronwall induction directly on the exact discrete recursion (not an assumption, not a check at more concrete $K$); independently re-derived from scratch and stress-tested by a dedicated hostile referee, verdict

[1] As of 2026-08-22, this general-$K$ proof is **unconditional**: the regularity/existence hypothesis it originally carried has since been proved, so the rate, the fixed-$K$ bridge for every $K$ (in particular $K \geq 11$), and the full statement $\varphi(n, c) \to \varphi_\infty(c)$ for every $c$ are all now theorems, not conditional statements. See the note after this abstract and Addenda 1, 2, 3 below for the precise statements and citations; the sentence this footnote is attached to, and the rest of this abstract, are left exactly as originally written.

**SOUND – WITH NAMED ISSUES** (four issues found, none fatal; two required correction and were fixed via dated addenda in the source document, following the same discipline used here). Consequently Theorem 2, Open Lemma 4, and Proposition 5 below are all now **unconditional** – for every $K \geq 0$ and every $c \geq 0$ respectively. The original conditional statements below are left exactly as written, since they were correct and correctly scoped at the time of writing; Addenda 1, 2, and 3, placed immediately after each, record the update and its citation without altering the original text. Archive record: `k2_open_lemma/k3_attempt_2/k6_attempt/k_general_existence_attempt/ATTEMPT.md` and its `adversarial/REFEREE_REPORT.md`.

# 1 The model

Fix $n \in \mathbb{N}$ and $c \geq 0$. Let $\pi$ be a uniformly random permutation of $[n]$. Independently, for each $i \in [n]$ let $\xi_i$ be i.i.d. Bernoulli with $P(\xi_i = 1) = c/n$, and let $U_i$ be i.i.d. uniform on $[n]$, independent of $\pi$ and of the $\xi$'s. Define the random mapping

$$f(i) = \begin{cases} U_i & \xi_i = 1 \quad (\text{“}i\text{ is rerouted”}) \\ \pi(i) & \xi_i = 0. \end{cases}$$

A point $i$ is *cyclic* for $f$ iff $f^t(i) = i$ for some $t \geq 1$, equivalently iff $i$ lies on a directed cycle of the functional digraph $i \mapsto f(i)$. The observable of interest is

$$\varphi(n, c) \;:=\; \frac{1}{n} E\big[\#\{i : i \text{ cyclic for } f\}\big] \;=\; P(1 \text{ is cyclic for } f)$$

(the second equality by exchangeability in $i$). $c$ is the expected number of rerouted points; $c = 0$ recovers a plain uniform permutation, so $\varphi(n, 0) \equiv 1$.

No name for this exact ensemble was located in the random-mapping literature (see §6): the closest studied family – Jaworski [6], Jaworski [7], Hansen–Jaworski [2, 4, 5] – is parametrized by an in-degree sequence, not by an independently corrupted background permutation.

# 2 The limit object and the main theorem

The $n \to \infty$ scaling limit of the cycle structure of a uniform random permutation is the Poisson–Dirichlet object PD(1) [8], with size-biased (“stick-breaking”) representation GEM(1), due to McCloskey (1965) and Patil–Taillie (1977); see Pitman [11], Ch. 3, for the construction and full attribution. Let $L(c)$ be this object equipped with an independent rate-$c$ Poisson process of marks on $[0, 1]$, each carrying an independent uniform destination: an unmarked point advances to the next point on its cycle; a marked point moves instead to its destination. This can be built explicitly from independent Exp(1) “closure clocks” per arc-head raced against the mark process, with the identification of this construction as the standard exploration representation of PD(1) under lazy revelation cited from [8, 1] and used as the single external structural input; every computation below is self-contained given that one citation. Write $\varphi_\infty(c) := P(x_0 \text{ cyclic in } L(c))$ for a uniform reference point $x_0$; by Fubini and exchangeability this equals $E[\text{Leb}(\text{cyclic set})]$, the natural limit of $\varphi(n, c)$.

**Theorem 1** (Closed form)**.** *For every $c \geq 0$,*

$$\varphi_\infty(c) \;=\; \int_0^1 e^{-ct^2}\, dt \;=\; \tfrac{1}{2}\sqrt{\pi/c}\, \operatorname{erf}(\sqrt{c}),$$

*with value* 1 *at* $c = 0$ *by continuity.*

*Proof sketch.* Reveal the forward orbit of $x_0$ step by step; let $t \in [0, 1)$ be the traversed fraction of mass. The orbit decomposes into *arcs* separated by surviving marks. At traversal level $t$: the $\pi$-closure hazard onto a specific open arc-head is $1/(1-t)$ per unit mass (the classical fact that the size-biased cycle length of a uniform permutation is Unif$(0, 1)$ is the $c = 0$, single-arc-head case of this); a mark at position $s < t$ kills (lands on visited territory) with probability $s$ or survives with probability $1 - s$, opening a new arc-head with its own closure clock.

Write $T_0 \sim$ Unif$(0, 1)$ for $x_0$'s own closure clock (an elementary fact about Exp(1) primitives), independent of the mark process. Fix $T_0 = t$. For a single surviving mark at $s < t$, the joint probability (over its independent survival Bernoulli and its independent closure clock) that it neither kills *nor* produces a sibling arc-head that closes before $t$ is

$$\underbrace{(1-s)}_{P(\text{survives})} \cdot \underbrace{\frac{1-t}{1-s}}_{P(\text{sibling closes after } t|\text{survives})} = 1-t,$$

**independent of** $s$ – the survival factor $(1-s)$ exactly cancels the competing-clock tail $(1-t)/(1-s)$. This is the entire mechanism behind the closed form. By the Poisson marking/thinning theorem, the number of "failing" marks in $[0, t)$ is then Poisson$(ct \cdot t)$ = Poisson$(ct^2)$ (rate $c$ on an interval of length $t$, constant fail-probability $t$, independent of position), so

$$P(x_0 \text{ cyclic} \mid T_0 = t) = P(\text{Poisson}(ct^2) = 0) = e^{-ct^2}.$$

Integrating over $T_0 \sim$ Unif$(0, 1)$ gives $\varphi_\infty(c) = \int_0^1 e^{-ct^2} dt$; the closed form follows from $u = \sqrt{c}\,t$ and the definition of erf. ☐

*Remark.* A heuristic tail computation that forgets to size-bias "the arc currently being traversed" – treating the inter-mark gaps as unconditional Exp$(c)$ rather than accounting for the inspection/waiting-time paradox – lands on $\sqrt{\pi/2}\, c^{-1/2}$ instead of the correct $(\sqrt{\pi}/2)\, c^{-1/2}$: a factor $\sqrt{2}$ error. The proof above never forms this quantity: the per-mark probability is computed jointly over the pair (survival Bernoulli, closure-clock exponential), never as an averaged arc length, so there is no size-biased object anywhere in the argument to get wrong.

**Corollary 1** (Series)**.** $\varphi_\infty(c) = \sum_{k\geq 0} \frac{(-c)^k}{k!(2k+1)}$ *for every real* $c$ *(entire function; proved by the Weierstrass* $M$*-test justifying term-by-term integration of* $e^{-ct^2}$*'s Taylor series on* $[0, 1]$*). First terms:* $1 - \frac{c}{3} + \frac{c^2}{10} - \frac{c^3}{42} + \cdots$*; in particular* $a_1 = 1/3$*.*

**Corollary 2** (Tail asymptotic, with a rigorous error bound)**.** *As* $c \to \infty$*,*

$$\varphi_\infty(c) = \frac{\sqrt{\pi}}{2}\, c^{-1/2} - R(c), \qquad 0 < R(c) < \frac{e^{-c}}{2c} \;\; \forall c > 0,$$

*so* $\varphi_\infty(c) = \frac{\sqrt{\pi}}{2} c^{-1/2}(1 + O(e^{-c}))$ *– a pure power-law tail up to exponentially small corrections. Proved by one integration by parts on the* erf *tail integral.*

## 3 The conditional-$K$ law and the Hansen–Jaworski connection

Condition on exactly $K$ marks rather than Poisson$(c)$ many.

**Lemma 1** (Mean, proved for every $K$)**.**

$$\varphi_K \;:=\; \int_0^1 (1-t^2)^K\,dt \;=\; \frac{4^K (K!)^2}{(2K+1)!}$$

*(a Wallis integral). Checks:* $\varphi_0 = 1$*,* $\varphi_1 = 2/3$*,* $\varphi_2 = 8/15$*,* $\varphi_3 = 16/35$*. Mixing over* $K \sim \mathrm{Poisson}(c)$ *exactly reproduces Theorem 1:* $\sum_K e^{-c}c^K/K! \cdot \varphi_K = \int_0^1 e^{-c}e^{c(1-t^2)}dt = \int_0^1 e^{-ct^2}dt$*.*

**Lemma 2** (Density at $K = 1$, proved)**.** *The cyclic-mass random variable* $M_1 := \mathrm{Leb}($*cyclic set*$)$ *has density* $f_{M_1}(x) = 2x$ *on* $(0,1)$*.*

*Proof sketch.* With a single reroute at a uniform position, the size-biased struck cycle has length $L \sim \mathrm{Unif}(0,1)$ (classical) and a uniform destination $u$. If $u \notin$ the struck cycle (probability $1-L$), the cyclic mass is deterministically $1-L$; if $u$ lands in the struck cycle (probability $L$), the cyclic mass is $1-L+D$ with $D \mid L \sim \mathrm{Unif}(0,L)$. Summing the two branches' contributions to the density of $M_1$ by the standard mixture change-of-variables formula gives $x + x = 2x$ on $(0,1)$. □

**Conjecture 1** (General-$K$ density, **not proved here**)**.** $f_{M_K}(x) = 2Kx(1-x^2)^{K-1}$ *on* $(0,1)$*, for* $K \geq 2$*.*

This reduces to Lemma 2 at $K = 1$; its mean matches $\varphi_K$ exactly (proved, by integration by parts – necessary but not sufficient for the density itself to be correct); it is supported by Kolmogorov–Smirnov tests at $K = 1,2,3$ (no rejection, two independent implementations) and by an external theorem for a structurally different microscopic model with the same conditional-$K$ limit:

> **Hansen & Jaworski [5], Theorem 7(ii).** For their model $\hat{T}_n^r$ – a uniform random mapping with $r$ vertices constrained to in-degree $\leq 1$ and $a = n - r$ vertices allowed in-degree $\leq 2$ – fixed $a$ and $k = \lfloor xn \rfloor$: $\Pr\{\hat{X}_n^r = k\} \sim \frac{1}{n} \cdot 2ax(1-x^2)^{a-1}$.

Identical functional form, with $a \leftrightarrow K$; the two microscopic models are structurally unrelated (their model is a uniform draw under an in-degree constraint, with no permutation-plus-reroute mechanism). This coincidence of the conditional-$K$ limit law is external supporting evidence, not a proof, of Conjecture 1 for this ensemble. Literature priority for this specific citation: *already known* – see §6.

## 4 The finite-$n$ bridge

Theorem 1 and Lemma 1 concern $L(c)$ itself, taken as given. Whether $\varphi(n,c) \to \varphi_\infty(c)$ as $n \to \infty$ is a separate question, only partially resolved.

For every finite $n$, writing $\varphi_n^{(K)}$ for $\varphi(n,c)$ conditioned on exactly $K$ rerouted points (a purely combinatorial quantity, independent of $c$, by exchangeability),

$$\varphi(n,c) \;=\; \sum_{K=0}^{n} \binom{n}{K} \Big(\frac{c}{n}\Big)^K \Big(1 - \frac{c}{n}\Big)^{n-K} \varphi_n^{(K)} \tag{1}$$

is an *exact* identity, not an approximation.

**Proposition 1** (Mixing reduction, proved unconditionally)**.** *If* $\varphi_n^{(K)} \to \varphi_K$ *for every fixed* $K \geq 0$*, then* $\varphi(n,c) \to \varphi_\infty(c)$*.*

*Proof sketch.* Split $\varphi(n,c) - \varphi_\infty(c)$ into a term measuring Binomial$(n, c/n)$ vs. Poisson$(c)$ total-variation distance (which $\to 0$ by Scheffé's lemma [12] applied to the elementary pointwise Poisson-limit computation – a quantitative rate $d_{TV} \le c^2/n$ is also available via Le Cam's bound [10], though only the qualitative limit is needed here) and a term measuring $\varphi_n^{(K)} - \varphi_K$ averaged over $K$, controlled uniformly in $n$ by a Chernoff-type tail bound on Binomial$(n, c/n)$ derived from scratch. Both vanish as $n \to \infty$. □

$K = 0$ is trivial and exact for every $n$ ($f = \pi$, a bijection, every point cyclic: $\varphi_n^{(0)} = 1$).

**Proposition 2** ($K = 1$ bridge, proved exactly)**.** *For every $n \ge 1$, $\varphi_n^{(1)} = \dfrac{2}{3} + \dfrac{1}{3n^2}$, exactly.*

*Proof sketch.* Fix the rerouted index; $\pi$ is then an unconditioned uniform permutation, and the cycle $C$ containing it has length $L$, exactly uniform on $\{1, \dots, n\}$ (a classical exact combinatorial fact, not merely a limit). Points outside $C$ remain cyclic regardless of the reroute's target $U$. A case split on $U$ inside $C$ (missing $C$ entirely; landing on the reroute source itself; landing $d$ steps into $C$) gives an exact count of cyclic points within $C$ given $L = \ell$; averaging over $L \sim$ Unif$\{1, \dots, n\}$ and $U$ gives the stated closed form. □

**Proposition 3** ($K = 2$ bridge, proved exactly)**.** *For every $n \ge 3$,*

$$\varphi_n^{(2)} \;=\; \frac{16n^3 + n^2 + 21n + 6}{30n^3} \;=\; \frac{8}{15} + \frac{1}{30n} + \frac{7}{10n^2} + \frac{1}{5n^3},$$

*exactly. In particular $\varphi_n^{(2)} \to \varphi_2 = 8/15$ as $n \to \infty$, at exact rate $\Theta(1/n)$ (leading term $1/(30n)$) –* not *$\Theta(1/n^2)$, the rate the $K = 0, 1$ pattern alone might suggest.*

*Proof sketch.* By exchangeability, split $\varphi_n^{(2)}$ into a weighted average of $\psi_n^{(2)}$ (the probability a *generic*, non-rerouted point is cyclic) and $\psi_n^{(2),R}$ (the same probability for one of the two rerouted points itself): $\varphi_n^{(2)} = \frac{2}{n}\psi_n^{(2),R} + \big(1 - \frac{2}{n}\big)\psi_n^{(2)}$, exact for every $n > 2$ – a reduction that holds, in this weighted-average form, for every fixed $K$, and by itself already shows that only the generic-point limit $\psi_n^{(K)} \to \varphi_K$ is needed to close the fixed-$K$ bridge at any $K$ (the rerouted-point term's weight $K/n \to 0$ regardless of $\psi_n^{(K),R}$'s own behaviour). For $K = 2$: tracing the forward orbit of a generic reference point as a discrete exploration process and case-splitting on whether each of the two rerouted sources lies on the reference point's own $\pi$-cycle – using the classical exact fact that a fixed point's $\pi$-cycle length is Unif$\{1, \dots, n\}$, plus the elementary fact that two fixed points of a uniform permutation of size $m$ share a cycle with probability exactly $1/2$, independent of $m$ – gives the exact closed form $\psi_n^{(2)} = 8/15 + 4/(15n) + 1/(15n^2)$; the parallel computation starting the walk at a rerouted point itself gives $\psi_n^{(2),R} = (5n + 2)(n + 1)/(12n^2)$. Recombining via the weighted average above gives the stated closed form for $\varphi_n^{(2)}$; its $1/n$ and $1/n^2$ terms arise from an incomplete cancellation between $\psi_n^{(2)}$'s and $\psi_n^{(2),R}$'s own $\Theta(1/n)$ corrections – the same mechanism that, at $K = 1$, cancels *completely* and produces $\varphi_n^{(1)}$'s cleaner $\Theta(1/n^2)$ rate. □

A single reroute disturbs exactly one background $\pi$-cycle (tractable, the mechanism behind Proposition 2); two reroutes already require tracking whether they strike the same cycle or different ones, and in what order, but the case count stays small enough for an explicit closed form (Proposition 3). Extending this hand case-analysis literally to $K = 3$ does grow combinatorially, exactly as expected – but a structurally different technique closes $K = 3$ anyway (and, mechanically, $K = 4, 5$ beyond it): the discrete exploration walk behind Propositions 2–3 is reformulated as an explicit $K$-*uniform* Markov chain on a 3-integer state, whose exact transition rule is derived once, for general $K$, from the same lazy-revelation fact used above; solving that chain in closed form is then a mechanical telescoping-sum computation, not a fresh case analysis for each $K$.

**Proposition 4** ($K = 3$ bridge, proved exactly, by a $K$-uniform transfer-matrix method)**.** *For every* $n \geq 4$,

$$\psi_n^{(3)} = \frac{16}{35} + \frac{12}{35n} + \frac{5}{28n^2} + \frac{3}{70n^3},$$
$$\varphi_n^{(3)} = \frac{16}{35} + \frac{1}{14n} + \frac{11}{10n^2} + \frac{23}{35n^3} + \frac{6}{35n^4},$$

*exactly, where* $\psi_n^{(3)}$ *is the "generic reference point" quantity of the* $K$*-general reduction underlying Proposition 3. In particular* $\varphi_n^{(3)} \to \varphi_3 = 16/35$ *as* $n \to \infty$*, at rate* $\Theta(1/n)$ *(leading term* $1/(14n)$*) – the same order as* $K = 2$*, not the* $\Theta(1/n^2)$ *of* $K = 1$*.*

*Proof sketch.* Track the forward-orbit exploration of a generic reference point by a state $(a, b, r)$: $a$ = points already queried as $\pi$-images (removed from the pool of future $\pi$-targets), $b$ = points reached by a reroute jump into previously unexplored territory (still eligible $\pi$-targets), $r$ = how many of the $K$ reroute sources remain unreached by the walk so far. The exact transition rule out of $(a, b, r)$ is elementary – the same permutation-exchangeability fact used in Proposition 3 – and is uniform in $K$: only the initial value $r = K$ changes with $K$. Solving the resulting linear recursion level by level in $r$ (a standard falling-factorial/hockey-stick telescoping identity, executed once per level, $r = 0, 1, 2, 3$) yields $\psi_n^{(3)}$ in closed form; the same weighted-average reduction as Proposition 3 (with $K = 3$ in place of $K = 2$) then gives $\varphi_n^{(3)}$. The method first reproduces the already-proved $\psi_n^{(1)}, \psi_n^{(2)}$ closed forms exactly, as an internal check, before being pushed one level further to $K = 3$; the $K = 3$ result is independently confirmed by exact rational brute-force enumeration through $n = 9$ (a fresh data point, not previously computed) and by a separately coded, non-symbolic recursion (archive record: `k2_open_lemma/k3_attempt_2/ATTEMPT.md`). □

$K = 4, 5$**: further closed forms, same mechanical method.** Because the procedure of Proposition 4's proof is uniform in $K$, climbing two further levels of the same recursion costs no new idea, only more arithmetic:

$$\psi_n^{(4)} = \frac{128}{315} + \frac{128}{315n} + \frac{103}{315n^2} + \frac{52}{315n^3} + \frac{4}{105n^4},$$
$$\psi_n^{(5)} = \frac{1024n^5 + 1280n^4 + 1405n^3 + 1105n^2 + 538n + 120}{2772n^5},$$

both exact, both verified against fresh brute-force enumeration, and both proving $\varphi_n^{(K)} \to \varphi_K$ unconditionally at $K = 4, 5$ via the same reduction identity as above. All of $K = 3, 4, 5$ were further re-derived, from the primitives up, by a separate adversarial referee who solved the recursion by a different technique (an integrating-factor method rather than the symbolic telescoping sum), substituted every closed form back into the recursion symbolically, re-ran every script from scratch, and reported no error at any layer (archive record: `k2_open_lemma/k3_attempt_2/adversarial/REFEREE_REPORT.md`).

$K = 6, \ldots, 10$**: the same mechanical method, run five further levels, unconditionally proved.** Running the identical procedure five rungs past $K = 5$ costs, again, no new idea:

$$\psi_n^{(6)} = \frac{2048n^6 + 3072n^5 + 4293n^4 + 4638n^3 + 3529n^2 + 1662n + 360}{6006n^6},$$

proving $\varphi_n^{(6)} \to \varphi_6 = 1024/3003$ unconditionally, via the same reduction identity, for every $n \geq 7$. The analogous exact closed forms for $\psi_n^{(7)}, \psi_n^{(8)}, \psi_n^{(9)}, \psi_n^{(10)}$ exist by the identical mechanical procedure and are each independently verified to have $n \to \infty$ limit exactly $\varphi_K$ (full formulas: archive record `k2_open_lemma/k3_attempt_2/k6_attempt/ATTEMPT.md` §1.1), extending the fixed-$K$ bridge's unconditional resolution through $K = 10$. The $\psi_n^{(6)}$ closed form above was independently re-verified by a separate hostile adversarial referee, who substituted every one of its 13 constituent closed forms (and, at $K = 7$, all 16 of that level's) back into the exact defining recursion with zero symbolic discrepancy in any case, and who matched the formula bit-for-bit against a fresh, independently-coded brute-force enumeration at $K = 6, n = 7$ (355081/823543, out of $7! \times 7^6 =$ 592,950,960 exhaustively enumerated combinations) and, as a second held-out point, at $K = 6, n = 8$ (191647/458752, out of $8! \times 8^6 =$ 10,569,646,080 combinations). The rate conjecture below was likewise confirmed exactly at each of $K = 6, \ldots, 10$ (same archive record, §1.1, §3.4), extending the pattern to eleven consecutive values, $K = 0, \ldots, 10$, with zero exceptions.

**Theorem 2** (General-$K$ rate, proved conditional on a precisely named regularity hypothesis)**.** $\lim_{n\to\infty} n\big(\psi_n^{(K)} - \varphi_K\big) = \dfrac{K}{4}\varphi_K$ *for every* $K \geq 0$. *This holds* ***unconditionally, with no hypothesis,*** *for* $K = 0, 1, \ldots, 10$*, read off directly, as an exact identity, from each of the eleven independently derived closed forms above (not a curve fit or an extrapolation). For* every $K$*, it is* ***proved*** *by the continuum scaling-limit argument sketched below,* ***conditional on*** *Hypothesis 1.*

**Proof sketch (the continuum technique, general $K$).** Reformulate the transfer-matrix recursion's state $(a, b, r)$ in the scaling variable $t = m/n$ and take the $n \to \infty$ limit *before* solving in $r$ – the reverse order from Proposition 4 onward, which solves the exact recursion level by level in $r$ first and only afterwards lets $n \to \infty$. Positing an expansion $g_r(m, b) = F_r(t, b) + \frac{1}{n}G_r(t, b) + O(1/n^2)$ turns the exact discrete recursion into a linear first-order ODE in $t$ at each order, with $r$ now an ordinary symbolic parameter rather than a summation index – sidestepping, by changing the order of operations, exactly the obstruction named below. Solving both ODEs by diagonal coefficient matching gives, for every $r$ symbolically,

$$F_r(t, b) = \sum_{k=0}^{r} \frac{r!}{(r-k)!} \cdot \frac{t^k}{\prod_{i=1}^{k+1}(r+b+i)}, \qquad G_r(t, b) = \sum_{k=0}^{r-1} \binom{k+2}{2} \frac{r!}{(r-k-1)!} \cdot \frac{t^k}{\prod_{i=1}^{k+2}(r+b+i)},$$

each proved to satisfy its defining recursion by an exact symbolic identity check for general $r, k, b$ (not curve-fitting). $F_r(1, 0) = \varphi_r$ for every $r$ independently re-derives the Wallis-integral mean of Lemma 1 by an entirely new route; $G_r(1, 0) = r\varphi_r/4$ for every $r$ – proved by an elementary binomial-sum symmetry identity, $\sum_{i=0}^{r-1}(r-i)(r-i+1)\binom{2r+1}{i} = r \cdot 2^{2r-1}$ – is exactly the rate claimed above.

**Hypothesis 1** (The one caveat, stated precisely)**.** *For every* $r$*,* $g_r(m, b)$ *admits a regular two-term asymptotic expansion* $F_r(t, b) + \frac{1}{n}G_r(t, b) + O(1/n^2)$*, of the specific polynomial-in-*$t$ *shape that the recursion's own algebra forces at each order. The* existence *of this expansion, for* $r$ *beyond the eleven concretely computed values* $K = 0, \ldots, 10$*, is not independently re-derived here from first principles (e.g. by a discrete-Gronwall-type error bound uniform in* $m, n$*).*

Wherever Hypothesis 1 can be checked against an independent, unconditionally proved computation, it holds exactly, with zero exceptions: $F_r$ against the Wallis integral $\varphi_K$ for $K = 0, \ldots, 6$ at $b = 0$ and against the full exact discrete formula, all $b$, for $K \leq 5$; $G_r$ against the rate for $K = 0, \ldots, 10$ at $b = 0$ and against the full exact discrete formula, all $b$, for $K \leq 5$. An independent hostile adversarial referee re-derived both ODEs by hand, the $F_r/G_r$ closed forms symbolically for general $r, k, b$, and the binomial-sum identity above, finding zero errors; the referee additionally tested $F_r, G_r$ against exact ground truth at $t \neq 1$ – 45 new data points, $r = 0, \ldots, 5$, symbolic $b$, off the single $t = 1$ point this document's own checks are otherwise confined to – finding zero discrepancies (new evidence for the ansatz within the checked range, not a proof for general $K$). Asked explicitly to judge whether Hypothesis 1 is correctly scoped, too conservative, or too optimistic, the referee's verdict, adopted here in full, is that **it is correctly scoped – neither too optimistic nor too conservative** (archive record: `k2_open_lemma/k3_attempt_2/k6_attempt/adversarial/REFEREE_REPORT.md`).

The induction-on-$r$ route named as a candidate strategy above has thus been executed – not on the exact discrete recursion itself, which remains unsolved symbolically in $r$ (see §7), but on its $n \to \infty$ scaling limit, taken *before* rather than after solving in $r$. This closes the general-$K$ rate question, conditionally.

A finite limit $\lim_n n(\psi_n^{(K)} - \varphi_K)$ forces $\psi_n^{(K)} - \varphi_K \to 0$ as an elementary, immediate consequence (if the difference did not vanish, multiplying by $n \to \infty$ could not converge to a finite value). So Theorem 2, exactly as proved – conditional on Hypothesis 1 – already entails the weaker statement $\psi_n^{(K)} \to \varphi_K$ conditionally, for every $K$, as an immediate corollary, not a separately established fact. The Open Lemma below is still stated, correctly, as **unconditionally** open for $K \geq 11$: Hypothesis 1's existence question, for $r$ beyond the eleven concretely checked values, is not independently resolved from first principles by anything in this document, so nothing here closes the bridge *unconditionally*. But it would be inconsistent to credit Theorem 2 with a conditional proof of the rate while denying that the same hypothesis conditionally settles the (logically weaker) bridge too – so both are recorded as resting on the identical, single, precisely named hypothesis, with the same status: proved conditionally, open unconditionally.

*Addendum* 1 (Hypothesis 1 is closed, 2026-08-22). Hypothesis 1 above is now **proved**, unconditionally, for every $r \geq 0$: a discrete-Gronwall induction on $r$, applied directly to the exact discrete recursion underlying $g_r, h_r$ (not an assumption, not a self-consistency check at more concrete $K$), shows the residual $R_r(m, b, n) := g_r(m, b) - F_r(t, b) - \frac{1}{n} G_r(t, b)$ is bounded by an explicit $D_r(b)/n^2$, uniformly over the entire domain of $m$ including the recursion's own base-case boundary – exactly the existence question Hypothesis 1 left open. Consequently **Theorem 2 now holds unconditionally, with no hypothesis, for every $K \geq 0$**, superseding the "conditional on Hypothesis 1" qualification in its statement above. Independently re-verified by a dedicated hostile referee (own simulator, own closed forms, $477 + 309$ exact identity checks, symbolic verification of the underlying ODEs for fully symbolic $r, k, b$, fresh probes at $r = 6, 7, 9, 10$ and $n$ up to $10^6$ never touched by the original proof); verdict **SOUND – WITH NAMED ISSUES** (four issues, none fatal, two requiring correction: an exponent typo in the bound's written justification, and a downstream $\Theta(1/n)$ overclaim at $K = 1$ in a related document, both corrected via dated addenda in the source, not silently). Archive record: `k_general_existence_attempt/ATTEMPT.md` and `k_general_existence_attempt/adversarial/REFEREE_REPORT.md`. A related, separately-derived fact goes further than the bare rate for $\psi_n^{(K)}$: combining this closure with Proposition 1-style reduction gives an *exact* formula for the $1/n$ coefficient of $\varphi_n^{(K)} - \varphi_K$ itself, for every $K \geq 1$: $\varphi_n^{(K)} - \varphi_K = K[\varphi_K/4 + F_{K-1}(1, 1) - \varphi_K]/n + O(1/n^2)$ – exactly zero at $K = 1$ (consistent with Proposition 2's $\Theta(1/n^2)$ rate) and verified strictly positive for $2 \leq K \leq 12$ (reproducing $1/30$ at $K = 2$ and $1/14$ at $K = 3$ from Propositions 3–4 above, and $1093/6006$ at $K = 6$), but **not proved positive for** $K \geq 13$ – so $\Theta(1/n)$, as opposed to the now-proved $O(1/n)$, remains unconditional only through $K = 12$; this narrower question is untouched by the closure above and should not be conflated with it.

**Open Lemma** (Fixed-$K$ bridge, $K \geq 11$)**.** $\lim_{n\to\infty} \varphi_n^{(K)} = \varphi_K$ *for every fixed integer* $K \geq 11$. ***Not proved or disproved unconditionally here****; proved conditional on Hypothesis 1, as an immediate corollary of Theorem 2 above.*

*Addendum* 2 (Open Lemma 4 is closed, 2026-08-22)*.* By Addendum 1, Hypothesis 1 is proved, so Theorem 2 holds unconditionally for every $K$; since a finite limit of $n(\psi_n^{(K)} - \varphi_K)$ elementarily forces $\psi_n^{(K)} - \varphi_K \to 0$, Open Lemma 4 above is now **PROVED, unconditionally, for every fixed integer** $K \geq 0$ (not just $K \geq 11$; combined with the unconditional Propositions 2–4 and the $K = 4, \ldots, 10$ closed forms of §4, this resolves the fixed-$K$ bridge for *every* $K$). It is no longer an open lemma; the name and label are kept for continuity with the rest of this document. The exact, all-orders, general-$K$ finite-$n$ closed form for $\psi_n^{(K)}$ (§7) is untouched by this closure and remains open.

**Proposition 5** (Conditional convergence)**.** *For every fixed* $c \geq 0$*,* $\varphi(n, c) \to \varphi_\infty(c)$ *as* $n \to \infty$*,* ***conditional on*** *Open Lemma 4 holding for every* $K \geq 11$ *– which Theorem 2 already discharges conditionally on Hypothesis 1, though not unconditionally.*

This follows immediately from Proposition 1 plus the unconditionally proved fixed-$K$ bridge for $K = 0, \ldots, 10$ (Propositions 2–4 together with the $K = 4, \ldots, 10$ closed forms above), given the open lemma. It is stated as a conditional proposition, not a theorem, because the open lemma is exactly what is missing; the empirical control that exists (§5) is evidence for, not a substitute for, resolving it. (Proposition 1's hypothesis – fixed-$K$ convergence for *every* $K \geq 0$ – is now discharged unconditionally through $K = 10$; the open lemma's scope has strictly narrowed from $K \geq 6$ to $K \geq 11$.)

*Addendum* 3 (Proposition 5 is now a theorem, 2026-08-22)*.* By Addendum 2, Open Lemma 4 is proved unconditionally for every $K$, so Proposition 1's hypothesis is discharged unconditionally for every fixed $K \geq 0$, and Proposition 5 above is no longer conditional:

*Theorem* 3 (Full bridge, now unconditional)*. For every fixed* $c \geq 0$*,* $\varphi(n, c) \to \varphi_\infty(c)$ *as* $n \to \infty$*,* ***unconditionally****, with no unproved hypothesis remaining.*

This is exactly the statement Proposition 5 above required the open lemma for; the archive records this promotion as "Proposição Condicional 5" becoming "Teorema 3." Not claimed here: a locally-uniform-in-$c$ (rather than pointwise-in-$c$) version of this convergence, which remains open (§7), and the exact all-orders closed form for $\psi_n^{(K)}$, likewise untouched.

# 5 Numerics

A companion package (`tamesis-cycle-survival/`, clean-room implementation, `numpy`/`scipy` only, fixed seeds) cross-checks every closed form above:

- **Series vs. closed form vs. independent quadrature** (`simulations/asymptotics.py`): agree to $< 10^{-9}$ across $c \in [0.01, 10]$ for the series (limited by double-precision catastrophic cancellation of the alternating series beyond $c \sim 15$, a floating-point artifact, not a defect of the series itself) and to $< 10^{-8}$ for erf-vs-quadrature up to $c = 500$.

- **Tail bound** (Corollary 2): the rigorous bound $0 < R(c) < e^{-c}/(2c)$ holds at every tested $c \in \{1, 2, 5, 10, 25, 50, 100, 500\}$ (at $c \geq 50$ the bound is below double-precision resolution for $\varphi_\infty(c)$'s magnitude and $R(c)$ correctly rounds to zero).

- **Exact finite-$n$ enumeration** (`simulations/finite_n.py`, no sampling): Proposition 2's closed form $\varphi_n^{(1)} = 2/3 + 1/(3n^2)$ verified as an exact rational identity by brute-force enumeration for $n = 1, \ldots, 9$. The same script's raw $K = 2$ table ($n = 2, \ldots, 8$) is exactly the data that first showed $\varphi_n^{(2)}$ converging monotonically toward $\varphi_2 = 8/15$ without the rescaled deviation $n^2(\varphi_n^{(2)} - \varphi_2)$ ever settling to a constant – now understood, via Proposition 3, as an artifact of rescaling by the wrong power of $n$ (the true rate is $\Theta(1/n)$, so $n^2 \cdot \mathrm{dev}$ necessarily grows without bound, while $n \cdot \mathrm{dev} \to 1/30$ only slowly). Proposition 3's closed form itself was verified, independently of this package, by exact rational brute-force enumeration up to $n = 9$ and by an adversarial from-scratch re-derivation (archive record: the theorem directory's `k2_open_lemma/` subfolder, files `ATTEMPT.md` and `adversarial/REFEREE_REPORT.md`); the full exact mixture (1) for $n = 2, \ldots, 4$ shows $|\varphi(n, c) - \varphi_\infty(c)|$ shrinking monotonically in $n$ at fixed $c \in \{0.5, 1, 2\}$. Proposition 4's $\psi_n^{(3)}$ and $\varphi_n^{(3)}$, and the $K = 4, 5$ closed forms of §4, were likewise verified outside this package by fresh exact brute-force enumeration (including a new $n = 9$, $K = 3$ data point never previously computed in the archive) and by a second, independent adversarial referee pass; neither check is reproduced by this package's own scripts, which stop at exact enumeration through the $K = 2$ case (archive record: `k2_open_lemma/k3_attempt_2/ATTEMPT.md` and `k2_open_lemma/k3_attempt_2/adversarial/REFEREE_REPORT.md`).

- **Monte Carlo at large $n$** (`simulations/monte_carlo.py`, $n = 20000$, 300 trials/cell, seed `20260822`): all tested $c \in \{0.1, 0.5, 1, 2, 5, 10, 30\}$ within $|z| < 4$ of $\varphi_\infty(c)$.

- **Poisson-mixture consistency**: $\sum_K \mathrm{Poisson}(K; c)\, \varphi_K$ reproduces $\varphi_\infty(c)$ to $< 10^{-8}$ for every tested $c$, confirming Lemma 1's consistency remark numerically.

A pytest suite of 49 checks (`tests/test_formula.py`) runs all of the above at small scale in under two seconds; see the package `README.md` for exact reproduction commands for every number in this paper.

## 6 Literature priority

A dedicated search (22 queries) traced the closest published family – Jaworski [6, 7], Hansen–Jaworski [2, 3, 4, 5] – and reached a verdict per component of this result:

- *The model* (§1): not found under this construction; the closest family is parametrized by in-degree sequence, not permutation-plus-independent-reroute.

- *The limit object* (PD(1) + Poisson marks): each ingredient is classical; the specific combination is not built this way in the source literature, which never starts from a permutation.

- *The conditional-$K$ law* (§3): **already known** – Hansen & Jaworski [5], Theorem 7(ii), for a structurally different model.

- *The Poisson-mixture closed form* (Theorem 1): not found; the strongest candidate component of this result for being previously unpublished. The mixture is a question that arises naturally only from the permutation-plus-reroute side (where #reroutes $\sim \mathrm{Binomial}(n, c/n) \to \mathrm{Poisson}(c)$ is immediate), so its absence from the in-degree-sequence literature is consistent with, not independent evidence against, its being previously unpublished.

- *The tail exponent* $1/2$: universal across the whole random-mapping literature (e.g. the classical #cyclic points $\sim \sqrt{\pi n/2}$ for a uniformly random mapping); the exact coefficient $\sqrt{\pi}/2$ and the purely exponential correction structure of Corollary 2 were not found.

This search is broad but not exhaustive: one classical source (Kolchin, *Random Mappings* [9]) remains unverified for lack of full-text access, and is reported here as inaccessible, not as ruled out. Every claim of priority in this paper is qualified as "not found in the searches performed," never as "does not exist" or "first."

# 7 Open problems

1. **Open Lemma 4** itself: the fixed-$K$ finite-$n$ bridge for $K \geq 11$ (resolved unconditionally at $K = 0, 1, \ldots, 10$ by Propositions 2–4 and the $K = 4, \ldots, 10$ closed forms of §4). This is the single hypothesis separating Proposition 5 from a full theorem. A continuum scaling-limit route (§4) now gives a complete, adversarially-verified conditional proof of the associated *rate* for every $K$ (Theorem 2), which – being a finite-limit statement, hence forcing $\psi_n^{(K)} \to \varphi_K$ as an elementary corollary – conditionally discharges the Open Lemma itself too, on the identical hypothesis. But Hypothesis 1 – existence of a two-term asymptotic expansion, for $K$ beyond the eleven concretely checked values – is not independently established from first principles, so the Open Lemma remains open *unconditionally* for $K \geq 11$, exactly as the rate does; the exact all-orders finite-$n$ closed form for general $K$ (of which §4's two-order expansion is only a truncation) is not attempted at all. **[Closed, 2026-08-22 – see Addenda 1–3.]** Hypothesis 1 is now proved unconditionally, so this item is resolved: Open Lemma 4 holds for every $K \geq 0$, and Proposition 5 is now Theorem 3, unconditional. Only the exact all-orders closed form for general $K$ named above remains open, untouched by this closure (see the new item below).

2. **The rate** of that bridge: now known exactly and unconditionally through $K = 10$ – $K = 0$ is exact (rate 0), $K = 1$ is exact at rate $1/(3n^2)$, and $K = 2, \ldots, 10$ are exact at $\Theta(1/n)$ (§5, Propositions 3–4 and the $K = 4, \ldots, 10$ closed forms above). **Theorem 2** proves the general-$K$ leading coefficient $K\varphi_K/4$ for every $K$, conditional on Hypothesis 1; unconditionally, with no hypothesis, it is confirmed only through $K = 10$. The $K = 1 \to K = 2$ jump from $\Theta(1/n^2)$ to $\Theta(1/n)$ remains a standing caution against assuming any rate extrapolates without a proof or a verified closed form. **[Closed, 2026-08-22 – see Addendum 1.]** Theorem 2's leading coefficient $K\varphi_K/4$ is now proved unconditionally for every $K$, not just through $K = 10$; and the *exact* $1/n$ coefficient of $\varphi_n^{(K)} - \varphi_K$ (not merely $\psi_n^{(K)} - \varphi_K$) is now known unconditionally for every $K \geq 1$ (Addendum 1), reproducing the $K = 1$ cancellation above as an exact-zero special case rather than a separate observation.

3. **Strict positivity, for $K \geq 13$,** of the exact $1/n$ coefficient of $\varphi_n^{(K)} - \varphi_K$ named in Addendum 1 – verified for $2 \leq K \leq 12$ but not proved in general. If positive for all $K \geq 2$, the rate is exactly $\Theta(1/n)$ there, not merely the now-proved $O(1/n)$; this is a narrower, separate question from the previous item's closure and is not resolved by it.

4. The growth rate, in $r$, of the error constants $D_r(b), C_r(b)$ in the discrete-Gronwall proof of Addendum 1 – the recursion defining them is explicit and terminating, but no closed-form or asymptotic growth rate was established; the numerics underlying that proof show the true error is far smaller than the bound in every case checked.

5. **Conjecture 1** (general-$K$ density) and its Poisson-mixture consequence, the full unconditional law $M(c) \stackrel{d}{=} \min(1, \sqrt{E/c})$, $E \sim \mathrm{Exp}(1)$ – a distributional claim unrelated to, and unresolved by, the fixed-$K$ bridge results above (those concern only the *mean* $\varphi_n^{(K)} \to \varphi_K$, not the full density $f_{M_K}$); not affected by the 2026-08-22 closure above.

6. A locally-uniform-in-$c$ (not merely pointwise) version of Theorem 3 (formerly Proposition 5, now unconditional – see Addendum 3 – but only pointwise in $c$, not locally uniform).

7. Any second-moment, concentration, or CLT statement for the cyclic fraction of either $\varphi(n, c)$ or $L(c)$ – entirely untouched here.

8. A first-principles (non-cited) proof of the exploration-process representation of PD(1) used as the single external input in §2.

9. **The exact, all-orders finite-$n$ closed form** for $\psi_n^{(K)}$ at general $K$ (of which §4's two-order expansion, now unconditional, is only a truncation) – separate, harder, and explicitly not attempted by any result above.

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